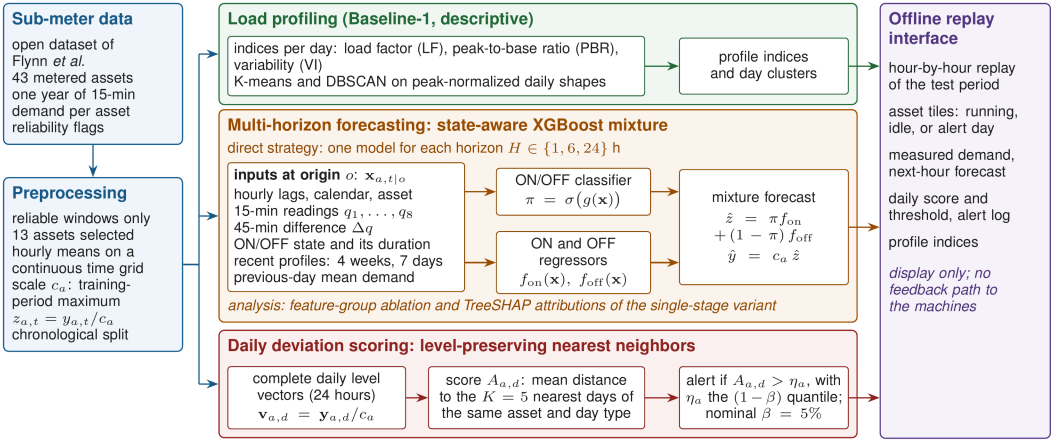

Sub-meter data
open dataset of Flynn et al.
43 metered assets
one year of 15-min demand per asset
reliability flags
Preprocessing
reliable windows only
13 assets selected
hourly means on a continuous time grid
scale c_a: training-period maximum
$z_{a,t} = y_{a,t}/c_a$
chronological split
Load profiling (Baseline-1, descriptive)
indices per day: load factor (LF), peak-to-base ratio (PBR), variability (VI)
K-means and DBSCAN on peak-normalized daily shapes
profile indices and day clusters
Multi-horizon forecasting: state-aware XGBoost mixture
direct strategy: one model for each horizon $H \in \{1, 6, 24\}$ h
inputs at origin o: $\mathbf{x}_{a,t|o}$
hourly lags, calendar, asset
15-min readings $q_1, \ldots, q_8$
45-min difference Δq
ON/OFF state and its duration
recent profiles: 4 weeks, 7 days
previous-day mean demand
ON/OFF classifier
$\pi = \sigma(g(\mathbf{x}))$
ON and OFF regressors
$f_{\text{on}}(\mathbf{x}),\ f_{\text{off}}(\mathbf{x})$
mixture forecast
$\hat{z} = \pi f_{\text{on}} + (1-\pi) f_{\text{off}}$
$\hat{y} = c_a \hat{z}$
analysis: feature-group ablation and TreeSHAP attributions of the single-stage variant
Daily deviation scoring: level-preserving nearest neighbors
complete daily level vectors (24 hours)
$\mathbf{v}_{a,d} = \mathbf{y}_{a,d}/c_a$
score $A_{a,d}$: mean distance to the $K = 5$ nearest days of the same asset and day type
alert if $A_{a,d} > \eta_a$, with η_a the $(1-\beta)$ quantile; nominal $\beta = 5\%$
Offline replay interface
hour-by-hour replay of the test period
asset tiles: running, idle, or alert day
measured demand, next-hour forecast
daily score and threshold, alert log
profile indices
display only; no feedback path to the machines